PRODUCT REQUIREMENTS DOCUMENT (PRD)

REQUISITION AND ISSUE SLIP (RIS) & INVENTORY MANAGEMENT PORTAL

APPENDIX 63 COMPLIANT

Document Version:	1.2.0 (Approved / Demo Release)	**Engineering Lead:**	System Architecture & Dev Team
Classification:	Sanitized Demo Portfolio	**Standard Benchmark:**	Government Accounting Manual App. 63
Last Updated:	October 2026	**Target Stack:**	React 19 + TypeScript + Supabase/Mock

PRIVACY & DATA DISCLAIMER (SANITIZED PORTFOLIO DEMO)

This repository contains a sanitized, open-source demo version of an application originally developed for a regional government agency. To comply with privacy requirements and non-disclosure standards, all sensitive production data, employee credentials, agency seals, and proprietary branding have been completely removed and replaced with sanitized mock records.

1. Executive Summary

The Requisition and Issue Slip (RIS) & Inventory Management Portal is a modern web application engineered to digitize, standardize, and accelerate the complete requisition, stock verification, multi-tier approval, and issuance lifecycle for public and enterprise institutions.

Originally built to replace slow, paper-driven routing adhering to statutory government standards (Government Accounting Manual - Appendix 63), this system eliminates physical document routing delays, prevents inventory discrepancy through atomic deduction algorithms, and provides audit-ready vector PDF generation and real-time consumption analytics.

- **Zero-Friction Requisitions**
  Composing multi-line requests with instant stock auto-completion in under 90 seconds.
- **Strict Separation of Duties**
  Enforced role guards between Requisitioners, Property Custodians, and Agency Directors.
- **Atomic Inventory Decrement**
  Real-time deduction upon approval prevents negative balances and stock discrepancies.
- **Appendix 63 PDF Generation**
  Statutory compliance with 4 official signature blocks rendered crisply on client-side.

1.1 Visual Requisition Lifecycle Overview

The state diagram below illustrates the complete lifecycle of a requisition, from initial composition to final archiving:

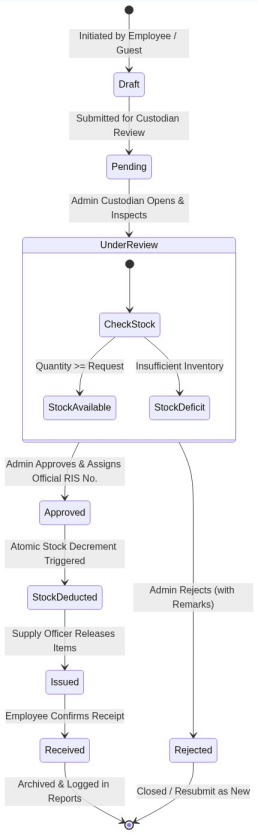


Figure 1.0: End-to-End Requisition Lifecycle & Stock Decrement State Machine

2. Problem Statement & Objectives

2.1 The Traditional Paper-Driven Problem

• Paper-Driven Routing Bottlenecks: Physical routing of slips across offices takes days, leading to misplaced files.
• Stock Blindness: Requisitioners request items with no live inventory visibility, causing high rejection rates.
• Inventory Inaccuracies: Manual ledger updates cause stock discrepancies, duplicate issuances, and shrinkage.
• Audit Tabulation Overhead: Compiling statutory quarterly reports requires manual, error-prone paper audits.

2.2 Core Product Objectives

• Requisition Composition < 2 min: Dynamic table builder with instant catalog autocomplete and unit autofill.
• Role-Based Guardrails: Distinct portals for Requisitioners, Property Custodians, and Super Admin Directors.
• Real-Time Atomic Stock Updates: Immediate decrement of inventory upon formal approval without race conditions.
• Statutory Document Fidelity: 100% compliant Government Appendix 63 printable/downloadable vector PDFs.
• Autonomous Analytics: Automated periodic consumption analytics (monthly, quarterly, semestral, annual).

3. Stakeholder & User Personas

The portal orchestrates four distinct roles with strict separation of duties and purpose-built dashboards:

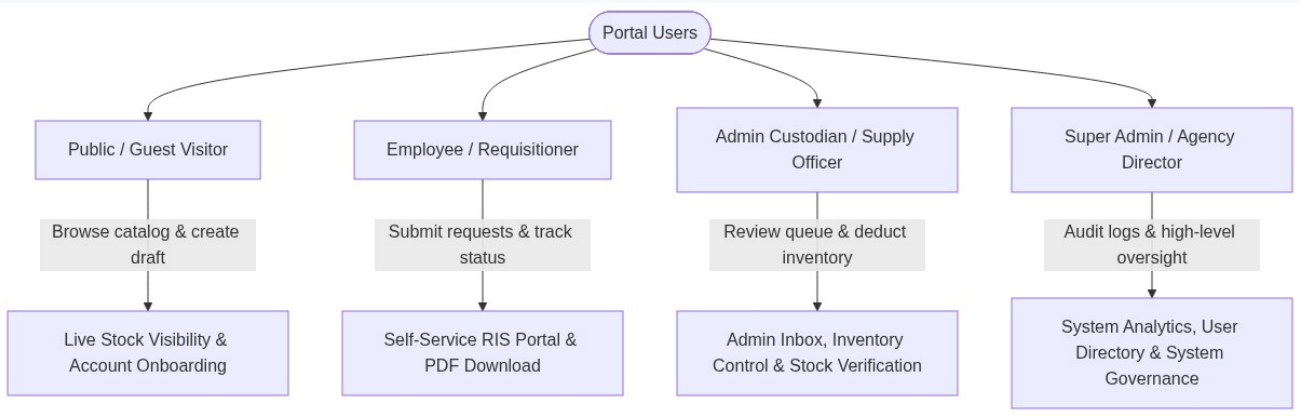


Figure 2.0: Stakeholder Roles, Portals & Architectural Interaction Pathways

3.1 Persona Specification Matrix

Role	Target Persona	Primary Responsibilities	Core Pain Points Addressed
Public / Guest	Unauthenticated Staff, New Hires, Evaluators	Browse catalog availability, compose draft requisition, register account.	Zero setup friction; transparent stock visibility before sign-in.
Employee (emp1, emp2)	Project Specialist, Research Analyst, Staff	Submit RIS, specify responsibility center & purpose, track approval, download signed PDFs.	Eliminates physical routing; transparent tracking of request approvals.
Admin Custodian (admin, admin2)	Property Custodian, Supply Officer	Review incoming requests, verify warehouse stock, assign official RIS No, approve/reject, adjust items.	Instant stock deduction; zero manual physical tallying; automated PDF generation.
Super Admin (superadmin)	Executive Director, Chief Information Officer	High-level audit across departments, system user governance, periodic consumption analytics.	One-click statutory reports; enterprise transparency; end-to-end auditability.

4. Functional Requirements & Feature Specifications

Module FR-1 **Requisition Formulation & Appendix 63 Engine**

- FR-1.1 Interactive Header: Captures Entity Name, Fund Cluster, Division, Office, Responsibility Center Code, and Purpose.
- FR-1.2 Dynamic Line Items: Supports real-time addition, deletion, and rearrangement of multiple supply line items.
- FR-1.3 Catalog Autocomplete: Typing item descriptions queries master inventory with auto-population of Stock No and Unit of Measure.
- FR-1.4 Live Stock Check: Alerts user immediately if requisitioned quantity exceeds current available warehouse balance.
- FR-1.5 Draft Persistence: Saves incomplete slips locally in browser storage with instant resume capability.

Module FR-2 **Approval Workflow & Central Review Inbox**

- FR-2.1 Administrative Review Queue: Filterable inbox categorizing slips by Draft, Pending, Approved, and Rejected states.
- FR-2.2 Stock Availability Verification: Custodian toggles stock availability (Yes/No) per item and adjusts approved quantity.
- FR-2.3 Custom RIS Assignment: Custodian assigns custom or auto-generated official RIS numbering (e.g., RIS-2026-001).
- FR-2.4 Rejection Reason Logging: Requires mandatory justification remarks when rejecting a requisition.
- FR-2.5 Atomic Stock Deduction: Automatic atomic stock decrement upon approval to prevent race conditions.

Module FR-3 **Inventory & Master Item Management**

- FR-3.1 Item Master Data: Records Stock No, Item Name, Detailed Specifications, Category (Equipment, Supplies), and Unit.
- FR-3.2 Real-Time Restock: Custodian can adjust stock quantities, modify specifications, or archive inactive items.
- FR-3.3 Search & Low Stock Alerts: Real-time filtering with visual badges highlighting items below threshold (< 5 units).
- FR-3.4 Batch Import Simulation: Built-in spreadsheet-like import preview for bulk inventory seeding.

Module FR-4 **Periodic Analytics & Reporting Engine**

- FR-4.1 Periodic Aggregation: Automated compilation by Month, Quarter, Semester, and Calendar Year.
- FR-4.2 Demand Ranking: Visualizes Top 5 Most Requested vs. Least Requested inventory supplies.
- FR-4.3 Departmental Distribution: Analyzes requisition consumption across administrative divisions.
- FR-4.4 Export Formats: Instant client-side export to Microsoft Excel (.xlsx) and printable PDF reports.

Module FR-5 **Official Appendix 63 PDF Generation Engine**

- FR-5.1 Statutory Appendix 63 Fidelity: Generates pixel-perfect PDFs matching statutory Government Accounting Manual standards.
- FR-5.2 Four Signature Blocks: Requested By (Employee), Approved By (Division Head), Issued By (Custodian), Received By (Officer).
- FR-5.3 Vector Table Pagination: Automatically paginates multi-line item slips with borders, remarks, and headers.

Module FR-6 **Multi-Role Access Control & Zero-Setup Demo Engine**

- FR-6.1 Zero-Setup Portability: Pre-seeded client-side store simulating full enterprise workflow without requiring cloud backends.
- FR-6.2 1-Click Role Switcher: Instant switcher between Employee, Admin Custodian, and Super Admin roles.
- FR-6.3 Data Reset Facility: One-click reset to restore clean sanitized state for continuous evaluations.

5. Non-Functional Requirements (NFR)

Key architectural pillars ensuring enterprise security, performance, responsive design, and compliance:

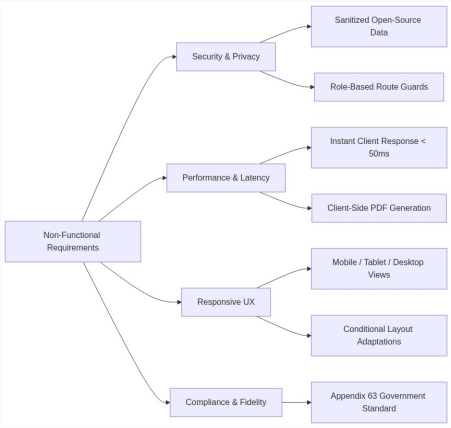


Figure 3.0: Non-Functional Architectural Pillars & Quality Attributes

- **Security & Privacy:** 100% sanitized data; strict role-based route guards; zero leakage of real agency identifiers.
- **Performance & Latency:** Client response < 50ms; instant client-side PDF compilation; sub-second initial load.
- **Responsive UX:** Optimized layouts for mobile phones, tablets/iPads, and enterprise desktop monitors.
- **Reliability & Integrity:** Atomic stock updates prevent negative stock counts; data validation prevents zero-item slips.

6. Success Metrics & Key Performance Indicators (KPIs)

Metric / Performance Indicator	Baseline Target	Demonstrated Demo Performance
Requisition Formulation Time	< 3 minutes per slip	**~90 seconds with live autocomplete**
Administrative Approval Velocity	80% faster than paper routing	**Instant digital review & 1-click approval**
Inventory Discrepancy Rate	0% variance	**0% discrepancy (automated atomic deduction)**
Appendix 63 PDF Generation	< 1.5 seconds latency	**~400ms (client-side jsPDF engine)**
Evaluator Onboarding Friction	Zero manual installation	**100% instant in-browser evaluation**

7. Assumptions & Operational Scope

• Demo Mode Execution: The application operates out-of-the-box utilizing an in-memory & LocalStorage mock engine, allowing immediate deployment to static CDNs (Vercel, Netlify, Cloud Run).

• Enterprise Backend Compatibility: Database schemas and data transfer objects are fully designed for zero-refactor binding to PostgreSQL 15+ and Supabase.

• Compliance Adherence: Slips and records are aligned with the statutory formats of the Philippine Government Accounting Manual (GAM) for National Government Agencies.

DOCUMENT APPROVAL & VERIFICATION BLOCK

PREPARED BY	REVIEWED BY	APPROVED FOR RELEASE
Lead Solutions Architect	**Senior Systems Auditor**	**Chief Technology Director**
Full-Stack Software Engineering	Compliance & Enterprise Quality	Public Sector Technology Practice